- observe the changes and draw conclusions about the properties of the lipids used in the above Experiment.

Key Answer

Water and oil	Water mixes into oil making a cloudy mixture with a thicker consistency. After 5-10 minutes the oil and water separate into two distinct layers with water on the bottom layer and oil on the top.
Adding detergent	Produces a cloudy mixture with a thin consistency and layer of foam on top. After 5–10 minutes this produces a liquid bottom layer and some foam above that indicating an emulsifier.
Adding sugar	Mixes into a thicker paste consistency then separates after 5–10 minutes into two separate layers.
Adding flour	Mixes into a milky substance with some oil globules visible in the liquid. After 5–10 minutes the solid does not dissolve but settles to the bottom, with two distinct layers of liquid and then oil as the top layer.
Adding mustard	This mixes in well producing a yellow liquid with powder particles visible on the walls of the bottle. After 5–10 minutes nothing has changed and the solution remains mixed well. This indicates to the student that the mustard is an emulsifier.
Adding egg white	Mixes into a cloudy gelatinous substance with no visible oil. After 5–10 minutes there are two (less distinct) layers. The bottom layer is a darker cream than the top with the top layer whiter and slightly foamy.
Adding egg yolk	This mixes in completely producing a cloudy yellow liquid that is still the same after 5–10 minutes, indicating an emulsifier.

SUMMARY

Lipids are a wide class of organic compounds present in living systems. They are a class of compounds soluble in non polar solvents. They have varied functions such as being energy reservoirs, acting as hormones, structural components of cell membranes etc., They also find wide applications in food industries, cosmetics, drug delivery systems etc. Lipids can be classified into simple lipids, compound lipids and derived lipids. Simple lipids are triglycerides and waxes. Triglycerides are esters of fatty acids with glycerol where as waxes are esters of fatty acids with other mono hydroxyl alcohols. Compound lipids are lipids with other chemical groups like phosphates in the case of phospholipids, sugars in the case of glycolipids and proteins in the case of lipoproteins. Derived lipids are the products of the hydrolysis of the first two classes of lipids such as fatty acids and cholesterol. Lipids have important properties like saponification, hydrogenation, rancidity etc which can be used to characterize a given lipid molecule.

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